

SDS8102: Advanced Mathematical Statistics

Lecture 7: Bayes Rules, Admissibility, and Minimaxity

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Lecture 7: Bayes Rules, Admissibility, and Minimaxity

Recall the statistical decision problem:

- observe $X \sim P_\theta$;
- choose an action a ;
- incur loss $L(\theta, a)$.

A decision rule δ is evaluated through its risk

$$R(\theta, \delta) = \mathbb{E}_\theta [L(\theta, \delta(X))].$$

But risk is a **function of θ** .

How should we compare two procedures when neither has smaller risk everywhere?

When Risk Curves Cross

Suppose two estimators δ_1 and δ_2 satisfy

$$R(\theta, \delta_1) < R(\theta, \delta_2)$$

for some values of θ , but

$$R(\theta, \delta_1) > R(\theta, \delta_2)$$

for others.

Then neither estimator dominates the other.

To choose between them, we need an additional principle.

Two fundamental approaches are:

Bayes	Minimax
average risk	worst-case risk

The Bayes Perspective

Suppose we assign a probability distribution π to the unknown parameter:

$$\theta \sim \pi.$$

The distribution π is called a **prior distribution**.

Instead of comparing the entire risk functions, we average risk with respect to π :

$$r(\pi, \delta) = \int_{\Theta} R(\theta, \delta) \pi(d\theta).$$

This is called the **Bayes risk** of δ under π .

A Bayes rule minimizes this average risk.

Definition: Bayes Rule

A decision rule δ_π is a Bayes rule with respect to π if

$$r(\pi, \delta_\pi) = \inf_{\delta} r(\pi, \delta).$$

The minimum value

$$r_\pi := r(\pi, \delta_\pi) = \inf_{\delta} r(\pi, \delta)$$

is called the **Bayes risk** of the statistical decision problem under π .

The prior converts a risk function into a single optimization criterion.

From Bayes Risk to Posterior Risk

The Bayes risk can be written as

$$r(\pi, \delta) = \mathbb{E} [L(\theta, \delta(X))],$$

where the expectation is over the joint distribution of (θ, X) .

Using iterated expectation,

$$r(\pi, \delta) = \mathbb{E} [\mathbb{E} [L(\theta, \delta(X)) \mid X]].$$

For an observed value $X = x$, define the **posterior risk**

$$\rho(a \mid x) = \mathbb{E} [L(\theta, a) \mid X = x].$$

Therefore Bayes risk can be minimized **pointwise in the data**.

Bayes Rule via Posterior Risk

For every observed x , choose

$$\delta_{\pi}(x) \in \arg \min_a \mathbb{E}[L(\theta, a) \mid X = x].$$

Why does this work?

For any decision rule δ ,

$$\mathbb{E}[L(\theta, \delta_{\pi}(X)) \mid X] \leq \mathbb{E}[L(\theta, \delta(X)) \mid X].$$

Taking expectation over X gives

$$r(\pi, \delta_{\pi}) \leq r(\pi, \delta).$$

Thus

Squared Error Loss

Suppose the action is an estimate of θ and

$$L(\theta, a) = (\theta - a)^2.$$

Given $X = x$, the posterior risk is

$$\rho(a | x) = \mathbb{E} [(\theta - a)^2 | X = x].$$

Using the bias–variance decomposition,

$$\rho(a | x) = \text{Var}(\theta | x) + (a - \mathbb{E}[\theta | x])^2.$$

Therefore the posterior risk is minimized at

$$\delta_{\pi}(x) = \mathbb{E}[\theta | X = x].$$

Under squared error loss, the Bayes estimator is the **posterior mean**.

Different Losses Give Different Bayes Rules

The Bayes rule depends critically on the loss function.

Loss	Bayes rule
$(\theta - a)^2$	Posterior mean
$ \theta - a $	Posterior median
$\mathbf{1}\{\theta \neq a\}$	Posterior mode

Thus there is no universally “best” Bayes estimator.

The optimal decision depends on how errors are penalized.

Example: Normal–Normal Model

Suppose

$$X \mid \theta \sim N(\theta, \sigma^2), \quad \theta \sim N(\mu_0, \tau^2).$$

Under squared error loss,

$$L(\theta, a) = (\theta - a)^2,$$

the Bayes estimator is the posterior mean.

By conjugacy,

$$\theta \mid X = x \sim N\left(\frac{\tau^2}{\tau^2 + \sigma^2}x + \frac{\sigma^2}{\tau^2 + \sigma^2}\mu_0, \frac{\sigma^2\tau^2}{\sigma^2 + \tau^2}\right).$$

Therefore

$$\delta_\pi(x) = \frac{\tau^2}{\tau^2 + \sigma^2}x + \frac{\sigma^2}{\tau^2 + \sigma^2}\mu_0.$$

Bayes Estimation as Shrinkage

Write

$$\delta_{\pi}(X) = \lambda X + (1 - \lambda)\mu_0, \quad \lambda = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Since

$$0 < \lambda < 1,$$

the Bayes estimator pulls X toward the prior mean μ_0 .

- Large τ^2 : weak prior information, so $\lambda \approx 1$ and $\delta_{\pi}(X) \approx X$.
- Small τ^2 : strong prior information, so $\lambda \approx 0$ and $\delta_{\pi}(X) \approx \mu_0$.
- Large σ^2 : noisy data produce stronger shrinkage toward μ_0 .

Thus Bayes estimation naturally produces a **shrinkage estimator**.

Risk of the Shrinkage Estimator

Consider

$$\delta_\lambda(X) = \lambda X + (1 - \lambda)\mu_0.$$

Since $X = \theta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$,

$$\delta_\lambda(X) - \theta = \lambda\varepsilon + (1 - \lambda)(\mu_0 - \theta).$$

Therefore

$$R(\theta, \delta_\lambda) = \lambda^2 \sigma^2 + (1 - \lambda)^2 (\theta - \mu_0)^2.$$

Compare this with the usual estimator $\delta_0(X) = X$:

$$R(\theta, X) = \sigma^2.$$

Shrinkage reduces variance but introduces bias.

Bayes Risk in the Normal–Normal Model

Average the risk over

$$\theta \sim N(\mu_0, \tau^2).$$

Since

$$\mathbb{E}_\pi[(\theta - \mu_0)^2] = \tau^2,$$

we obtain

$$r(\pi, \delta_\lambda) = \lambda^2 \sigma^2 + (1 - \lambda)^2 \tau^2.$$

Differentiating with respect to λ ,

$$2\lambda\sigma^2 - 2(1 - \lambda)\tau^2 = 0,$$

which gives

$$\boxed{\lambda^* = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Thus minimizing integrated frequentist risk gives exactly the same estimator as minimizing posterior expected loss.

From Bayes Optimality to Admissibility

Recall that δ is **inadmissible** if another rule δ' satisfies

$$R(\theta, \delta') \leq R(\theta, \delta) \quad \text{for every } \theta,$$

with strict inequality for at least one θ .

A natural question is:

If a rule is Bayes optimal, must it be admissible?

Not always without additional conditions.

But a particularly useful result holds when the Bayes rule is unique.

Unique Bayes Rules Are Admissible

Theorem

Bayes Admissibility Theorem

Suppose δ_π is a Bayes rule under prior π and has finite Bayes risk. If δ_π is the unique Bayes rule, then

δ_π is admissible.

Idea. Suppose δ_π were inadmissible. Then there would exist δ' such that

$$R(\theta, \delta') \leq R(\theta, \delta_\pi) \quad \forall \theta.$$

Integrating with respect to π gives

$$r(\pi, \delta') \leq r(\pi, \delta_\pi).$$

Since δ_π is Bayes, equality must hold. Hence δ' is also Bayes, contradicting uniqueness.

Why Uniqueness Matters

Suppose δ_π is Bayes but another rule δ' satisfies

$$R(\theta, \delta') \leq R(\theta, \delta_\pi) \quad \forall \theta,$$

with strict inequality only on a set A such that

$$\pi(A) = 0.$$

Then

$$r(\pi, \delta') = r(\pi, \delta_\pi).$$

Thus both procedures can have exactly the same Bayes risk even though δ' dominates δ_π .

The prior average may fail to detect improvements on sets receiving zero prior probability.

Bayes optimality alone does not automatically imply admissibility.

Another Route to Admissibility

Theorem

Bayes Admissibility via Full Support

Suppose δ_π is Bayes under a prior π . Assume:

- π has **full support**: every nonempty open set $U \subseteq \Theta$ satisfies $\pi(U) > 0$;
- $R(\theta, \delta)$ is continuous in θ for every δ .

Then δ_π is admissible.

Proof idea. If δ' dominates δ_π , then for some θ_0 ,

$$R(\theta_0, \delta') < R(\theta_0, \delta_\pi).$$

By continuity, the strict inequality holds on some neighborhood U of θ_0 . Since π has full support,

$$\pi(U) > 0.$$

Therefore, $r(\pi, \delta') < r(\pi, \delta_\pi)$, contradicting Bayes optimality.

What If We Do Not Want to Choose a Prior?

The Bayes approach evaluates a rule by its average risk

$$r(\pi, \delta) = \int R(\theta, \delta) \pi(d\theta).$$

But the answer depends on the chosen prior π .

Suppose instead that we want a procedure that performs reasonably well even at the **worst possible parameter value**.

For a decision rule δ , consider

$$\sup_{\theta \in \Theta} R(\theta, \delta).$$

This is the worst-case risk of δ .

The minimax principle chooses the rule with the smallest worst-case risk.

Minimax Decision Rule

Definition: Minimax Rule

A decision rule δ^* is minimax if

$$\sup_{\theta \in \Theta} R(\theta, \delta^*) = \inf_{\delta} \sup_{\theta \in \Theta} R(\theta, \delta).$$

The quantity

$$R^* = \inf_{\delta} \sup_{\theta \in \Theta} R(\theta, \delta)$$

is called the **minimax risk**.

The two optimization principles are therefore

Bayes: minimize average risk, Minimax: minimize worst-case risk.

A Simple Example

Suppose

$$X \sim N(\theta, 1), \quad L(\theta, a) = (\theta - a)^2.$$

Consider the usual estimator

$$\delta_0(X) = X.$$

Its risk is

$$R(\theta, \delta_0) = \mathbb{E}_\theta[(X - \theta)^2] = 1.$$

Thus

$$\sup_{\theta \in \mathbb{R}} R(\theta, \delta_0) = 1.$$

So the worst-case risk of X is exactly 1.

But how can we prove that no other estimator has worst-case risk strictly smaller than 1?

A Lower Bound from Bayes Risk

For any prior π and any decision rule δ ,

$$r(\pi, \delta) = \int R(\theta, \delta) \pi(d\theta) \leq \sup_{\theta} R(\theta, \delta).$$

Therefore

$$\inf_{\delta} r(\pi, \delta) \leq \inf_{\delta} \sup_{\theta} R(\theta, \delta).$$

Since

$$r_{\pi} = \inf_{\delta} r(\pi, \delta),$$

we obtain

$$\boxed{r_{\pi} \leq R^*}.$$

Thus **every Bayes risk provides a lower bound on the minimax risk.**

Using Bayes Rules to Prove Minimaxity

Suppose we find a rule δ^* and a prior π such that

$$\sup_{\theta} R(\theta, \delta^*) = r_{\pi}.$$

But we always have

$$r_{\pi} \leq R^* \leq \sup_{\theta} R(\theta, \delta^*).$$

Hence all three quantities must be equal:

$$r_{\pi} = R^* = \sup_{\theta} R(\theta, \delta^*).$$

Therefore δ^* is minimax.

This gives a powerful strategy:

Find a Bayes lower bound that matches the worst-case risk.

Example: Gaussian Mean

Return to

$$X \sim N(\theta, 1), \quad L(\theta, a) = (\theta - a)^2.$$

Consider the prior

$$\theta \sim N(0, \tau^2).$$

The Bayes estimator is

$$\delta_\tau(X) = \frac{\tau^2}{1 + \tau^2} X.$$

The posterior variance is

$$\text{Var}(\theta \mid X) = \frac{\tau^2}{1 + \tau^2}.$$

Under squared error loss, the Bayes risk is the expected posterior variance, so

$$r_\tau = \frac{\tau^2}{1 + \tau^2}.$$

Let the Prior Become Diffuse

As

$$\tau^2 \rightarrow \infty,$$

the Bayes risk satisfies

$$r_\tau = \frac{\tau^2}{1 + \tau^2} \rightarrow 1.$$

Since every Bayes risk is a lower bound on the minimax risk,

$$R^* \geq \sup_{\tau^2 > 0} r_\tau = 1.$$

But we have

$$\delta_0(X) = X \implies \sup_{\theta} R(\theta, \delta_0) = 1.$$

Therefore

$$1 \leq R^* \leq 1 \implies R^* = 1, \delta_0(X) = X \text{ is minimax.}$$

An Important Subtlety

Notice that X was **not** the Bayes rule for any finite τ^2 in our sequence.

Instead,

$$\delta_\tau(X) = \frac{\tau^2}{1 + \tau^2} X \longrightarrow X$$

as $\tau^2 \rightarrow \infty$, while

$$r_\tau \longrightarrow 1.$$

Thus a sequence of proper priors can be used to establish a minimax lower bound.

The limiting “flat prior”

$$\pi(\theta) \propto 1$$

is improper, so we do not need to treat it as an ordinary prior.

The proof uses only the proper priors

$$N(0, \tau^2), \quad \tau^2 < \infty.$$

How Hard Can a Prior Make the Problem?

For every prior π , the Bayes risk satisfies

$$r_\pi = \inf_{\delta} r(\pi, \delta) \leq \inf_{\delta} \sup_{\theta} R(\theta, \delta) = R^*.$$

Thus every prior gives a lower bound on the minimax risk.

A natural question is:

Which prior gives the largest Bayes risk?

This motivates

$$\sup_{\pi} r_{\pi}.$$

A prior that makes the Bayes problem as difficult as possible is called a **least favorable prior**.

Least Favorable Prior

Definition: Least Favorable Prior

A prior π^* is least favorable if

$$r_{\pi^*} = \sup_{\pi} r_{\pi}.$$

Recall that

$$r_{\pi} \leq R^* \quad \text{for every } \pi.$$

Hence

$$\sup_{\pi} r_{\pi} \leq R^*.$$

The least favorable prior chooses the distribution over parameters under which the best possible average performance is worst.

Bayes Rule + Constant Risk

There is an especially useful situation.

Suppose δ_π is Bayes under π and

$$R(\theta, \delta_\pi) = c \quad \text{for every } \theta.$$

Then

$$r_\pi = \int R(\theta, \delta_\pi) \pi(d\theta) = c.$$

Also,

$$\sup_{\theta} R(\theta, \delta_\pi) = c.$$

But

$$r_\pi \leq R^* \leq \sup_{\theta} R(\theta, \delta_\pi).$$

Therefore

$$\boxed{R^* = c.}$$

A More General Bayes–Minimax Theorem

Constant risk is sufficient, but not necessary.

Theorem

Bayes Rule is Minimax

Suppose δ_π is Bayes under π and

$$r_\pi = \sup_{\theta \in \Theta} R(\theta, \delta_\pi).$$

Then δ_π is minimax.

Indeed,

$$r_\pi \leq R^* \leq \sup_{\theta} R(\theta, \delta_\pi).$$

The two endpoints are equal, so

$$r_\pi = R^* = \sup_{\theta} R(\theta, \delta_\pi).$$

Why Does Equality Occur?

Since δ_π is Bayes,

$$r_\pi = \int R(\theta, \delta_\pi) \pi(d\theta).$$

But an average can equal the maximum only if the prior places its mass where the risk is maximal.

Thus, if

$$r_\pi = \sup_{\theta} R(\theta, \delta_\pi),$$

then

$$R(\theta, \delta_\pi) = \sup_{\vartheta} R(\vartheta, \delta_\pi)$$

for π -almost every θ .

A least favorable prior therefore tends to concentrate on the **hardest parameter values**.

Bayes and Minimax as a Game

Think of two players:

- the **statistician** chooses a decision rule δ ;
- nature chooses the parameter θ .

The statistician wants to minimize risk; nature wants to make it large.

The minimax problem is

$$\inf_{\delta} \sup_{\theta} R(\theta, \delta).$$

If nature is allowed to randomize over θ using a prior π , the corresponding problem becomes

$$\sup_{\pi} \inf_{\delta} r(\pi, \delta).$$

We always have

$$\sup_{\pi} \inf_{\delta} r(\pi, \delta) \leq \inf_{\delta} \sup_{\theta} R(\theta, \delta).$$

Under suitable conditions, equality holds.

Minimax Theorem

Under suitable regularity conditions,

$$\sup_{\pi} \inf_{\delta} r(\pi, \delta) = \inf_{\delta} \sup_{\theta} R(\theta, \delta).$$

Equivalently,

$$\text{maximum Bayes risk} = \text{minimax risk}.$$

When the supremum on the left is attained by π^* , π^* is least favorable.

Its Bayes rule is then a natural candidate for a minimax rule.

This gives a deep connection:

$$\text{Bayes} \longleftrightarrow \text{Minimax}$$

through least favorable priors.

Back to the Gaussian Mean Problem

Suppose

$$X \sim N_d(\theta, I_d), \quad \theta \in \mathbb{R}^d,$$

with squared error loss

$$L(\theta, a) = \|a - \theta\|^2.$$

Consider the usual estimator

$$\delta_0(X) = X.$$

Its risk is

$$R(\theta, \delta_0) = \mathbb{E}_\theta \|X - \theta\|^2 = d.$$

Therefore

$$\boxed{R(\theta, \delta_0) = d \quad \forall \theta.}$$

Thus δ_0 has constant risk.

Is X Minimax?

Take the prior

$$\theta \sim N_d(0, \tau^2 I_d).$$

Under squared error loss, the Bayes estimator is

$$\delta_\tau(X) = \frac{\tau^2}{1 + \tau^2} X.$$

The posterior covariance is

$$\text{Var}(\theta \mid X) = \frac{\tau^2}{1 + \tau^2} I_d.$$

Hence the Bayes risk is

$$r_\tau = \frac{d\tau^2}{1 + \tau^2}.$$

As $\tau^2 \rightarrow \infty$,

$$r_\tau \longrightarrow d.$$

X Is Minimax in Every Dimension

Since Bayes risk gives a lower bound on minimax risk,

$$R^* \geq \sup_{\tau^2 > 0} \frac{d\tau^2}{1 + \tau^2} = d.$$

On the other hand,

$$\sup_{\theta} R(\theta, X) = d,$$

so

$$R^* \leq d.$$

Therefore

$$\boxed{R^* = d.}$$

Hence

$$\boxed{\delta_0(X) = X \text{ is minimax for every } d \geq 1.}$$

Nothing special happens at $d = 3$ from the perspective of minimaxity.

But Recall the James–Stein Phenomenon

For $d \geq 3$, consider the James–Stein estimator

$$\delta_{\text{JS}}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right) X.$$

As we saw earlier,

$$R(\theta, \delta_{\text{JS}}) < R(\theta, X) = d \quad \forall \theta.$$

Therefore, for $d \geq 3$,

X is inadmissible.

But we just proved

X is minimax.

There is no contradiction.

Minimax Does Not Imply Admissible

The two notions ask different questions.

Admissibility : Can another rule improve risk everywhere?

Minimaxity : How small is the worst-case risk?

James–Stein improves upon X pointwise:

$$R(\theta, \delta_{JS}) < R(\theta, X).$$

Nevertheless,

$$\sup_{\theta} R(\theta, \delta_{JS}) = d.$$

Thus both estimators attain the minimax risk d .

A minimax estimator can be inadmissible.

Why Does James–Stein Remain Minimax?

For every θ ,

$$R(\theta, \delta_{\text{JS}}) < d.$$

But the improvement becomes arbitrarily small as $\|\theta\|$ grows:

$$R(\theta, \delta_{\text{JS}}) \longrightarrow d \quad \text{as } \|\theta\| \rightarrow \infty.$$

Hence

$$\sup_{\theta \in \mathbb{R}^d} R(\theta, \delta_{\text{JS}}) = d.$$

So James–Stein improves risk at every finite θ , but it does not improve the **supremum** risk.

Both X and δ_{JS} are therefore minimax, while only one dominates the other.

The Dimension Threshold

For the Gaussian mean problem,

$$X \sim N_d(\theta, I_d),$$

the estimator X behaves very differently depending on dimension:

	Admissible	Minimax
$d = 1, 2$	✓	✓
$d \geq 3$	×	✓

For $d \geq 3$, James–Stein shows that X is inadmissible.

But the diffuse-prior argument shows that its constant risk d is still the smallest possible worst-case risk.

This illustrates why we need **multiple notions of optimality**.

Three Notions of Optimality

We have now encountered three different criteria:

Criterion	Question
Admissible	Can another rule dominate it?
Bayes	Does it minimize average risk under π ?
Minimax	Does it minimize worst-case risk?

These notions are related, but they are not equivalent.

Important connections include

Bayes + suitable conditions $\implies$ admissible,

and

Bayes risk = maximum risk $\implies$ minimax.

A Finite Decision Problem

Suppose

$$\theta \in \{0, 1\},$$

and we must choose an action

$$a \in \{0, 1\}.$$

Consider two decision rules with risk table

	$\theta = 0$	$\theta = 1$
δ_1	0	1
δ_2	1/2	1/2

Thus

$$\sup_{\theta} R(\theta, \delta_1) = 1, \quad \sup_{\theta} R(\theta, \delta_2) = \frac{1}{2}.$$

δ_2 has the smaller worst-case risk.

Can we recover this conclusion using a Bayes argument?

Finding the Least Favorable Prior

Let

$$\pi(\theta = 0) = p, \quad \pi(\theta = 1) = 1 - p.$$

The Bayes risks of the two rules are

$$r(\pi, \delta_1) = 1 - p, \quad r(\pi, \delta_2) = \frac{1}{2}.$$

Hence the optimal Bayes risk is

$$r_\pi = \min \left\{ 1 - p, \frac{1}{2} \right\}.$$

As a function of p ,

$$r_\pi = \begin{cases} 1/2, & p \leq 1/2, \\ 1 - p, & p > 1/2. \end{cases}$$

Therefore

$$\sup_{\pi} r_{\pi} = \frac{1}{2}.$$

Matching the Bayes and Minimax Bounds

For δ_2 ,

$$R(0, \delta_2) = R(1, \delta_2) = \frac{1}{2}.$$

Therefore

$$\sup_{\theta} R(\theta, \delta_2) = \frac{1}{2}.$$

From the least favorable prior,

$$\sup_{\pi} r_{\pi} = \frac{1}{2}.$$

Hence

$$\sup_{\pi} r_{\pi} = R^* = \sup_{\theta} R(\theta, \delta_2) = \frac{1}{2}.$$

Thus δ_2 is minimax. This illustrates the general strategy:

Bayes lower bound = worst-case risk $\implies$ minimaxity.

Lecture 7: Takeaways

- Risk $R(\theta, \delta)$ evaluates a rule at each parameter value.
- A **Bayes rule** minimizes average risk under a prior π .
- Under squared error loss, the Bayes estimator is the posterior mean.
- Bayes estimators naturally produce **shrinkage**.
- Unique Bayes rules are admissible; full support and continuity give another route to admissibility.
- A **minimax rule** minimizes the worst-case risk.
- Every Bayes risk gives a lower bound on the minimax risk.
- Least favorable priors connect Bayes and minimax procedures.
- Minimavity does **not** imply admissibility.

For the Gaussian mean problem,

X is minimax for every d ,	X is inadmissible for $d \geq 3$.
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